

A top-down view of a grey desk with various medical and study items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a white coffee cup. A large white pill is visible on the left side of the central text box.

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CONSTIPATION

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

CONSTIPATION: ADULTS

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LEGEND

ADR Adverse Drug Reaction

BM Bowel Movements

GI Gastrointestinal

OIC Opioid-Induced Constipation

PEG Polyethylene Glycol 3350

IBS Irritable Bowel Syndrome

IBD Inflammatory Bowel Disease

Licensed to Shreyansh Salunke (ID: 4477)

LEARNING OBJECTIVES

- Define constipation in adults
- Understand how to assess and evaluate patients with constipation
- Know the red flags and when to refer patients
- Understand pharmacological and non-pharmacological treatment options

WHAT IS CONSTIPATION?

The definition of constipation **varies** among individuals and includes (not limited to the following:

Infrequent defecation (less than 3 times/week) along with other symptoms:

- Straining or painful defecation
- Hard, lumpy stools
- Incomplete evacuation
- Bloating and abdominal discomfort
- A sense of rectal blockage

Classifications of Primary Constipation

1. **Normal colonic transit**
 - Normal bowel motility rate + feeling of incomplete evacuation
2. **Slow transit constipation**
 - Delay in stool transit + infrequent defecation
3. **Pelvic floor dysfunction**
 - Ineffective defecation due to pelvic floor muscular dysfunction and/or coordination issues

ROME IV CRITERIA IN ADULTS

FYI

The Rome IV criteria for chronic functional constipation include the following:

- The following must be present for ≥ 3 months with onset of ≥ 6 months before diagnosis:
 - At least 2 of the following
 - Lumpy or hard stools for $> 25\%$ of defecations
 - Straining for $> 25\%$ of defecations
 - Sensation of incomplete evacuation in $> 25\%$ of defecations
 - Sensation of anorectal obstruction/blockage for $> 25\%$ of defecations
 - Manual maneuvers to assist $> 25\%$ of defecations (e.g. digital evacuation)
 - < 3 spontaneous defecations/week
 - No (or rare) loose stools without the use of laxatives
 - Insufficient criteria to consider Irritable Bowel Syndrome (IBS)

RISK FACTORS

Risk factors

- Female
- Pregnancy
- > 65 years of age
- Lower education or socioeconomic status
- Changes in diet or eating disorders
- Low caloric, fiber and/or fluid intake
- Living conditions (e.g., physical abuse)
- Sedentary lifestyle
- Ignoring the urge to defecate

Medical conditions

- CV disease: stroke
- GI conditions: irritable bowel syndrome, hemorrhoids
- Neurologic conditions: Parkinson's disease, spinal cord injuries
- Metabolic conditions: diabetes, thyroid disorders, $\downarrow\text{Mg}^{2+}$, $\uparrow\text{Ca}^{2+}$, $\downarrow\text{K}^{+}$
- Psychiatric conditions: depression, anxiety

MEDICATION CAUSES- SECONDARY CONSTIPATION

Anticholinergics

Antidepressants

Antidiarrheal drugs

Antiepileptics

Antihistamines

Antihypertensives

Antinauseants

Anti-Parkinson's
drugs

Antipsychotics

Antispasmodics

Cation-containing
agents (iron,
calcium, aluminum)

Diuretics

NSAIDs

Opioids

Vinca alkaloids

5-HT₃ antagonists

PATIENT ASSESSMENT

S Symptoms

C Characteristics

H History

O Onset

L Location

A Aggravating factors

R Remitting factors

H History of medical conditions

A Allergies

M Medications (including vitamins, herbals)

S Social history (e.g., pregnancy)

Assess for red flags
(next slide)

RED FLAGS

New onset in children < 2 or adults > 50

Family history of colon cancer

Recent abdominal surgery

Unexplained weight loss > 5%

Diagnosed IBS-C or IBD

Eating disorders

GI Bleeds

Vomiting, rash, fever, or extreme thirst

Persistent abdominal pain or
unremitting nocturnal symptoms

Abdominal or rectal mass

Unexpected acute changes in bowel
behaviour

Symptoms of iron deficiency with or
without anemia

Severe pain during defecation

Impaction



GOALS OF THERAPY

- ✓ Identify and treat underlying cause(s)
- ✓ Educate patients on appropriate laxative use
- ✓ Alleviate discomfort
- ✓ Establish normal bowel routine
- ✓ Prevent complications

NON-PHARMACOLOGICAL MEASURES

Dietary Modifications:

- Encourage sufficient daily fibre intake (25-35g)
- Promote adequate fluid intake (2.2-3L) to complement fiber intake

Behavioral Changes:

- Avoid straining to prevent complications
- Establish a normal bowel routine for the patient
- Eliminate the urge to avoid defecation
- Encourage regular physical activity

PHARMACOLOGICAL MEASURES

5-HT₄ Receptor agonists

Guanylate Cyclase-C agonists

Lavage solutions

μ-opioid receptor antagonists

Bulk-forming agents

Stool softeners

Lubricants

Hyperosmotic agents

Stimulant agents

Osmotic agents

PHARMACOLOGICAL MEASURES - NON-PRESCRIPTION

BULK FORMING -Psyllium Onset: 1-3 days	1 st Line	• Safe for chronic use (contraindicated in fecal impaction) • Take one dose with at least 8oz of water to prevent fecal impaction • Main ADR: Bloating, flatulence, abdominal discomfort (start low dose)
OSMOTIC -Polyethylene Glycol (PEG 3350) Onset 2-4 days	1 st Line	• More effective and more tolerable than lactulose • Effective in opioid-induced constipation • Separate from other medications by at least 2 hours • Safe for long-term use
OSMOTIC -Lactulose Onset 1-2 days		• Can be used in diabetes, although still monitor for hyperglycemia • Effective in opioid-induced constipation • Separate from other medications by at least 2 hours
OSMOTIC -Sorbitol 70% solution		• Equally as effective as lactulose (onset: 1-2 days) • Effective in opioid-induced constipation • Concern for bowel necrosis (when used concurrently with sodium polystyrene sulfonate)
OSMOTIC -Glycerin suppositories Quick onset (~15min – 30 min)		• If possible, to be retained for 15 min • Effective in opioid-induced constipation

PHARMACOLOGICAL MEASURES - NON-PRESCRIPTION

OSMOTIC [SALINE LAXATIVE] -Magnesium citrate, magnesium hydroxide sodium phosphate enema	• Rapid onset: < 30 minutes - 6 hours (enemas have an onset of < 15 minutes) • Contain electrolytes: avoid in renal failure/cardiac disease • Can cause electrolyte imbalances • Usually for bowel cleansing (ex: magnesium citrate)
STIMULANT -Senna, bisacodyl Onset: PO 6-12 h Bisacodyl PR: 0.5-1 h	• Effective in opioid-induced constipation • Potential for dependence • An option if osmotic laxatives are not tolerated/fail • Senokot leads to less abdominal discomfort
STOOL SOFTENERS/ EMOLLIENT -Docusate sodium, mineral oil Onset: 1-3 days	• Limited efficacy data, generally NOT recommended • Avoid mineral oil: risk of aspiration; can also bind to fat-soluble drugs
LAVAGE -Picosulfate sodium/magnesium oxide/citric Acid; electrolyte Solutions	• Used before a colonoscopy or medical procedure

PHARMACOLOGICAL MEASURES- PRESCRIPTION

5-HT₄ RECEPTOR AGONIST: -Prucalopride Onset: 24-48 h	• Officially indicated for adult female patients (>18) with chronic idiopathic constipation who have not responded to two previous classes of laxatives • Has GI prokinetic activity • Discontinue if ineffective after 4 weeks of treatment
μ-OPIOID RECEPTOR ANTAGONIST: - Methylnaltrexone: Onset 30 min to 4 h - Naloxegol: Onset 6 h	• Methylnaltrexone: opioid-induced constipation in palliative care patients; SC injection; adjunctive therapy to laxatives • Naloxegol: treatment of opioid-induced constipation (OIC) in adult patients with non-cancer pain who have not responded adequately to laxatives • Contraindicated with strong CYP3A4 inhibitors • Both are expensive and are reserved for refractory cases
GUANYLATE CYCLASE-C RECEPTOR AGONIST: -Linaclotide, plecanatide Onset: 24-48 h	• Indication: chronic idiopathic constipation and IBS-C in adults when traditional therapies (e.g. PEG, Bisacodyl) are ineffective • Increases intestinal fluid and decreases GI transit time but can cause diarrhea

PREGNANCY AND BREASTFEEDING

PREGNANCY

1. Encourage non-pharmacological options first
2. Bulk-forming agents with adequate hydration are first line therapy due to lack of systemic absorption
3. If no improvement: can add or switch to PEG or lactulose, but for a short duration
4. Glycerin suppositories or bisacodyl suppositories can be used for short-term

DO NOT USE IN PREGNANCY: mineral oil, castor oil, linaclotide, prucalopride

BREASTFEEDING

1. Encourage non-pharmacologic options first
2. Bulk-forming agents with adequate hydration are first line therapy due to lack of systemic absorption
3. Magnesium hydroxide considered second line
4. Bisacodyl & Senna may be used for short-term

DO NOT USE IN BREASTFEEDING: linaclotide, prucalopride

* Docusate should not be used due to lack of efficacy

TAKEAWAY

Use patient factors to choose treatment

Bulk-forming
laxatives are
contraindicated in
fecal impaction

Suppositories and
enemas have a
quicker onset of
action compared
to oral laxatives

Opioids should
always be **paired**
with either a
stimulant or
osmotic laxative

Patients can
become
dependent on
stimulant laxatives

STEP-WISE SUMMARY CHRONIC CONSTIPATION

1

Initiate non-pharmacological methods (ensure to add fiber to diet)
If inadequate fibre → consider bulk forming laxatives



2

If no relief in 4-6 weeks: Osmotic agents along with fibre supplementation
[PEG > Lactulose in terms of ADR]



3

If no relief in 4-8 weeks: Stimulant (short duration)



4

If no relief: Prescription drug therapy

COLONOSCOPY

FYI

- An examination of the rectum and colon using a colonoscope (a thin, flexible tube with a tiny camera on the end)
- Requires “bowel preparation”

Note: Detailed instructions provided by clinics, this is a general guide

- Clear liquid diet starting 1 day before procedure
 - Drink at least 4L of liquid/day (including any liquid medication for prep)
 - Variety of fluids to replenish electrolytes (not just water)
- Stop taking certain medications: iron, codeine, loperamide
- Take medication to empty colon
 - Bisacodyl
 - Polyethylene glycol (PEG)
 - Tips: refrigerate the night before to improve taste; drink ~1 cup/hour, if experiencing nausea, wait 20-30min then try again
 - Saline-based formulas (sodium phosphate)
 - Tablet form: alternative to drinking prep, may increase compliance
- **Caution in patients with heart failure or renal disease due to risk of electrolyte imbalance**
 - PEG formulas: considered safer option
 - Saline-based formulas: risk of electrolyte imbalance & mineral overdose

CASE 1

CP is a 48-year-old female who presents to her local pharmacy with concerns regarding constipation. CP normally has a bowel movement every other day, however over the last week she has only had two bowel movements. As a result, she is experiencing bloating and abdominal discomfort. She is seeking advice, especially because she will be going in for surgery in the next month and is worried her pain medications may make this worse. CP's past medical history is significant for osteoarthritis, hypertension, type 2 diabetes and depression. She is allergic to clindamycin, having developed a rash when she took it a few years ago. CP is currently taking nifedipine XL 30 mg PO daily, metformin 1000 mg PO daily, saxagliptin 5 mg PO daily, escitalopram 10 mg PO daily and Tylenol Arthritis® (acetaminophen) 1300 mg PO BID. She eats relatively well and exercises frequently.



CASE 1 - QUESTIONS

1. Which of the following is NOT one of CP's risk factors for developing constipation?
 - a) Diabetes
 - b) Depression
 - c) Age
 - d) Female sex
2. Which of the following is NOT an appropriate goal of therapy for CP?
 - a) Alleviate her bloating and abdominal discomfort
 - b) Daily bowel movements
 - c) Prevent complications
 - d) Educate patient on various non-pharmacological options
3. CP will be undergoing a right knee replacement in 1 month and will likely be discharged on a course of opioids for pain management. Which of the following would be the LEAST ideal agent to recommend for constipation prophylaxis?
 - a) Senna
 - b) Bisacodyl
 - c) Metamucil
 - d) PEG3350
4. Which of the following is FALSE?
 - a) In fecal-impacted patients, bulk-forming laxatives are the safest option
 - b) Stimulant laxatives are often recommended for opioid-induced constipation
 - c) Suppositories have a faster onset of action than stimulant laxatives
 - d) The onset of action of bulk-forming agents is approximately 1-3 days

CASE 2

SL is a 30-year-old female patient presenting to the pharmacy with worsening constipation over the past 10 days. She reports hard stools, straining with defecation, and a bowel movement frequency of twice per week. SL is 18 weeks pregnant. She works remotely as a hotel interior designer and was recently promoted to project manager for the construction of a new hotel in Toronto. This promotion has led to a significantly heavier workload and a more sedentary lifestyle. Her only regular physical activity is occasionally walking her dog during phone meetings. She denies abdominal pain, rectal bleeding or other concerning symptoms. At her most recent physician visit, SL was reported to be healthy and is currently only taking a daily prenatal multivitamin. Her water intake is typically two glasses per day, but she has recently increased this to three glasses daily without any improvement in her symptoms.



CASE 2- QUESTIONS

1. Which of the following is the most appropriate recommendation for SL?
 - a) Senna
 - b) Psyllium husk
 - c) Bisacodyl suppository
 - d) Mineral oil

2. SL returns after 6 days with minimal improvement. What is the most appropriate next step in managing her constipation?
 - a) Discontinue psyllium husk and switch to mineral oil
 - b) Continue psyllium husk and add PEG 3350
 - c) Switch to bisacodyl suppository daily for long-term use
 - d) Continue psyllium husk and add linaclootide

CONSTIPATION: CHILDREN

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CONSTIPATION IN CHILDREN

Phases of increased risk for constipation:

1. Introduction of solid food or cow's milk
2. Inadequate fiber and fluid intake; excessive milk consumption (slows intestinal motility)
3. Toilet training
 - Stool withholding
4. Beginning school or daycare
 - Stool withholding; changes in BM schedule; reluctance to use school/daycare toilets

Classified into two categories:

- **Functional constipation**
 - Most common; no organic etiology
 - Normal difficulty passing stool or infrequent stools that feel incomplete, without underlying autonomic or neurological
- **Organic constipation**
 - Seen in < 5% of children
 - Potential causes include anatomical abnormalities, medical conditions; requires referral

GOALS OF THERAPY

- ✓ Alleviate signs and symptoms of discomfort
- ✓ Establish normal bowel routine
- ✓ Prevent complications

ROME IV CRITERIA FOR DIAGNOSING FUNCTIONAL CONSTIPATION

FYI

Criteria for children < 4 years

- No suspicion of irritable bowel syndrome AND
- 1 month of at least 2 of the following:
 - < 3 defecations/week
 - History of excessive stool retention
 - History of painful or hard BMs
 - Presence of a large fecal mass in the rectum
 - History of large diameter BMs (which may obstruct the toilet in toilet-trained children)
 - At least 1 episode of incontinence/week (in toilet-trained children)

Criteria for children ≥4 years

- No suspicion of irritable bowel syndrome AND
- 2 of the following present weekly for at least 1 month prior to diagnosis:
 - < 3 defecations/week
 - At least one incident of fecal incontinence/week
 - History of painful or hard BMs
 - History of retentive posturing or stool retention
 - Large fecal mass in the rectum
 - History of large diameter BMs that may obstruct the toilet
 - Not explained by another medical condition after evaluation

RED FLAGS

- ☐ Inability to pass meconium within the first 48 hours
- ☐ Suspected or diagnosed organic constipation
- ☐ Medication-induced
- ☐ Rectal bleeding or blood in the stool
- ☐ Signs of infection
- ☐ Vomiting, fever
- ☐ Significant abdominal distention with pain
- ☐ Intermittent loose stools, ribbon stools
- ☐ Symptoms of nutritional deficiencies: fatigue, poor appetite, weight loss, lethargy
- ☐ Fecal impaction suspected or Explosive stool after a digital rectal exam
 - ☐ Mass in the abdominal region/"rabbit poop"
- ☐ Failure to thrive or developmental delays



NON-PHARMACOLOGICAL MEASURES

- Encourage adequate fluid intake per weight from water, fruits and juices
- Recommend adequate fiber intake ~5g/day in children ≤ 2
 - E.g. pureed vegetables, fiber-containing infant cereal, prunes, peas, fruits, etc.
- Promote daily physical activity
- Develop and optimize bowel routine by using a low stool or making a trip 30 minutes after a meal
- Avoid straining or sitting on the toilet for more than 5 minutes
- Keep a bowel movement diary
- Lookout for signs of withholding bowels

PHARMACOLOGICAL MEASURES

Recommendations for constipation in children are NOT evidence-based

PEG3350	1 st line; safe and commonly used
Lactulose	2 nd line; less effective than PEG3350; causes more bloating and abdominal pain
Glycerin suppositories	Safe and effective, good option if PO route refused/for immediate relief (particularly in infants)
Senna, bisacodyl	Refractory cases; used when osmotic laxatives fail (contraindicated in infants)
Mineral oil	Not recommended: increased risk of aspiration (lipoid pneumonia); less effective than PEG3350

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CHANGE LOG

August 2025

- Slide 15: Added more ADRs to psyllium, minor edit in amount of water intake
- Slides 32,33: New case and questions added

November 2025

- Some slides rearranged for better flow
- Slide 11: Updated red flags
- Removed questions 5 and 6 from case 1
- Minor grammar and wording changes throughout

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 15: Updated notes on sorbitol
- Slide 17: Added plecanatide
- Slide 18: Simplified, removed docusate from meds to avoid and added a separate note on it
- Updated references